

Double/Debiased Machine Learning for Treatment Analysis

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1 Introduction and Motivation

Research Question

The primary research question studied is how to obtain valid inferential statements about treatment effects in the presence of highly complex, high-dimensional confounding nuisance parameters. Specifically, this work addresses whether we can evaluate treatment effects efficiently to avoid the heavy biases that arise when naively plugging machine learning estimators of confounders into classical estimation problems.

Significance of the Problem

This problem is of significant interest because empirical researchers in the social sciences frequently face high-dimensional confounding variables where traditional classical parametric assumptions fail. In these highly complex spaces, estimating the effect of the confounder variables on our variables of interest is difficult, and traditional frameworks do not work very well. The use of machine learning methods is perfect to handle many covariate effects, but it is possible to encounter bias caused by regularization and overfitting. Therefore, in this paper, we analyze the application of the best practices to establish a robust method for removing these biases and drawing the right causal conclusions from modern complex data. This entire method is called "Double/Debiased Machine Learning".

Related Literature

This work builds upon two key areas of semiparametric research. The first involves $\sqrt{N}$ -consistent estimation using classical nonparametric estimators (e.g., Robinson, 1988; Newey, 1990), whose reliance on Donsker conditions makes them unsuitable for high-dimensional settings. The second focuses on targeted maximum likelihood learning (van der Laan and Rubin, 2006) and using sample-splitting to relax strict entropy conditions (Bickel, 1982; Belloni et al., 2012).

Methodological Contribution

This paper serves as an accessible toolbox for researchers approaching this topic for the first time, providing a clear pathway into more complicated applications. The main contribution is offering a general and simple procedure for estimating and conducting inference that remains formally valid in highly complex settings, by demonstrating that the impact of regularization bias and overfitting can be removed using two simple ingredients: Neyman-orthogonal moments and cross-fitting. Specifically, this work shows the application of the DML (Debiased Machine Learning) method to

the simple framework of Partially Linear Regression, estimating the ATE and CATE using simple machine learning methods: Lasso and Random Forest.

Paper Structure

The remainder of this paper is organized as follows. The next section details the methodology, introducing the Partially Linear Regression (PLR) model and the core steps of the Double/Debiased Machine Learning (DML) framework, including an extension to Conditional Average Treatment Effects (CATE). The subsequent section describes the 1991 SIPP dataset used for the empirical analysis, highlighting the key variables and the identification strategy. Finally, the last section presents and discusses the results, comparing the performance of Lasso and Random Forest in estimating the Average Treatment Effect (ATE) and evaluating the treatment impact across different subgroups.

2 Methodology

The Empirical Model and General Econometric Techniques

In modern empirical research, the estimation of a low-dimensional target parameter of interest, denoted as θ_0 , is often complicated by the presence of a nuisance parameter, denoted as η_0 . The target parameter is typically defined as the solution to a moment condition:

$$E[m(W; \theta_0, \eta_0)] = 0 \quad (1)$$

where W denotes the observed random variables and m is a known score function. The nuisance parameter η_0 is an auxiliary object that must be accounted for to identify the parameter of interest, but it is not the primary focus of the research. In many interesting contemporary applications, η_0 is high-dimensional and highly complex, representing objects such as conditional expectation functions.

To estimate these complex nuisance parameters without relying on restrictive and ad hoc parametric assumptions, researchers increasingly turn to flexible Machine Learning (ML) tools. A natural, naive approach would be to obtain a first-step ML estimate of the nuisance parameter, $\hat{\eta}$, and plug it into the sample analog of the moment condition to solve for the target estimator $\hat{\theta}$. However, this "plug-in" approach generally behaves poorly and leads to invalid inference because the estimation error in $\hat{\eta}$ propagates directly into the estimator $\hat{\theta}$. This distortion is driven by two distinct forms of sensitivity: regularization bias and overfitting bias.

Regularization bias occurs because high-dimensional ML methods rely on regularization (trading off variance for bias) to prevent their variance from exploding. Consequently, the estimation

error $(\hat{\eta} - \eta_0)$ is not centered at zero, causing a heavy bias in the estimator of θ_0 that prevents it from achieving $\sqrt{N}$ -consistency. Overfitting bias, on the other hand, is a more subtle issue that arises because the estimator $\hat{\eta}$ is a random function of the same data used to construct the sample moment condition. This statistical dependence generates large distortions and further invalidates standard asymptotic approximations.

To solve these problems, the Double/Debiased Machine Learning (DML) approach relies on two essential statistical ingredients: Neyman orthogonality and cross-fitting. Neyman orthogonality is a local robustness property of the score function, ensuring that the moment condition is insensitive to small local perturbations of the nuisance parameter around its true value. Cross-fitting is a form of repeated sample splitting that restores independence between the estimation error of the nuisance function and the data used to estimate the target parameter, thereby eliminating overfitting bias.

The Partially Linear Regression Model

A specific and highly relevant application of this general framework is the Partially Linear Regression (PLR) model. The PLR model is defined by the following system of equations:

$$\begin{aligned} Y &= D\theta_0 + g_0(X) + U, & E[U \mid X, D] &= 0 \\ D &= m_0(X) + V, & E[V \mid X] &= 0 \end{aligned} \quad (2)$$

where Y is the outcome variable, D is the policy or treatment variable of interest, X is a potentially high-dimensional vector of controls, and U and V are disturbances. In this model, $g_0(\cdot)$ represents the relationship between the controls and the outcome, while $m_0(\cdot)$ represents the confounding dependence of the treatment on the controls. The functions $g_0(\cdot)$ and $m_0(\cdot)$ are the nuisance parameters, and they can be as flexible as desired. The major drawback of this model is that it assumes θ is linear and equal for the entire population, but this is also why it is so simple and easy to understand.

If we apply a naive approach, we would construct a standard linear predictor estimator taking the form:

$$\hat{\theta}_0 = \left(\frac{1}{n} \sum_{i \in I} D_i^2 \right)^{-1} \frac{1}{n} \sum_{i \in I} D_i (Y_i - \hat{g}_0(X_i)) \quad (3)$$

where $\hat{g}_0$ is the ML estimator of the nuisance function. The scaled error of this naive estimator can be decomposed as $\sqrt{n}(\hat{\theta}_0 - \theta_0) = a + b$. While the term a converges to zero under mild conditions, the term b represents the regularization bias, which diverges.

$$b = (ED_i^2)^{-1} \frac{1}{\sqrt{n}} \sum_{i \in I} m_0(X_i) (g_0(X_i) - \hat{g}_0(X_i)) + o_P(1) \quad (4)$$

It diverges because the ML estimator $\hat{g}_0$ converges at a slower rate due to the regularization usually applied in ML methods; this bias term will generally explode, rendering the estimator invalid.

To overcome this, we rely on a partialling-out approach to create an orthogonalized estimator. By defining the orthogonalized regressor $V = D - m_0(X)$, we can partial out the effect of X from D . After estimating the residuals $\hat{V}$, we can plug them into the naive estimation and obtain:

$$\check{\theta}_0 = \left(\frac{1}{n} \sum_{i \in I} \hat{V}_i D_i \right)^{-1} \frac{1}{n} \sum_{i \in I} \hat{V}_i (Y_i - \hat{g}_0(X_i)) \quad (5)$$

This is the correct orthogonalized estimator.

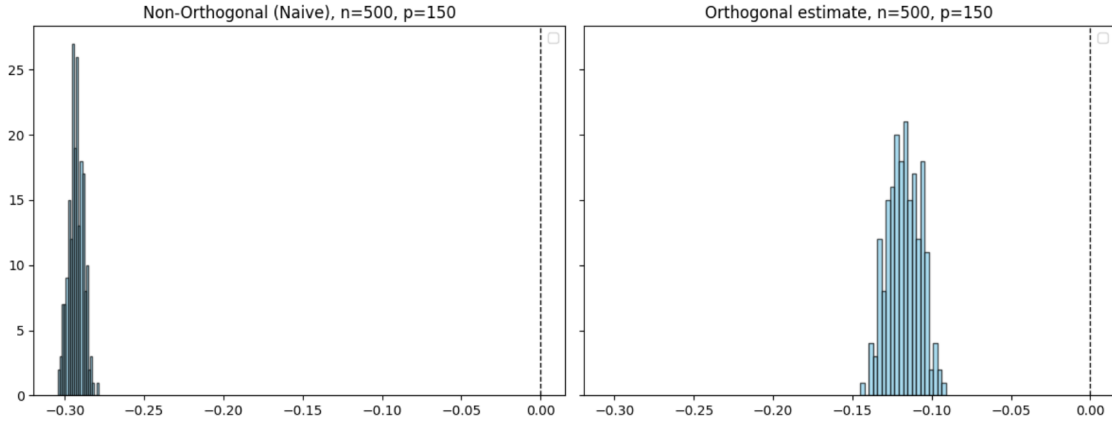


Figure 1: Effect of the Orthogonal estimator compared to the naive one, calculated on synthetic data. The line at 0 represents the perfect estimation.

This orthogonalization yields a new scaled error formulation, $\sqrt{n}(\tilde{\theta}_0 - \theta_0) = a^* + b^* + c^*$, where the new bias term b^* depends only on the product of the estimation errors of both nuisance functions, allowing it to vanish asymptotically. However, the c^* component needs more than orthogonal estimation in order to vanish.

$$c^* = \frac{1}{\sqrt{n}} \sum_{i \in I} V_i (\hat{g}_0(X_i) - g_0(X_i)) + \dots \quad (6)$$

This component is a normalized sum of products between structural unobservable errors and nuisance estimation errors, which are normally correlated due to overfitting. So, in order to avoid this, the estimates of the nuisance parameters are performed on one sample and the orthogonal estimation on a different sample.

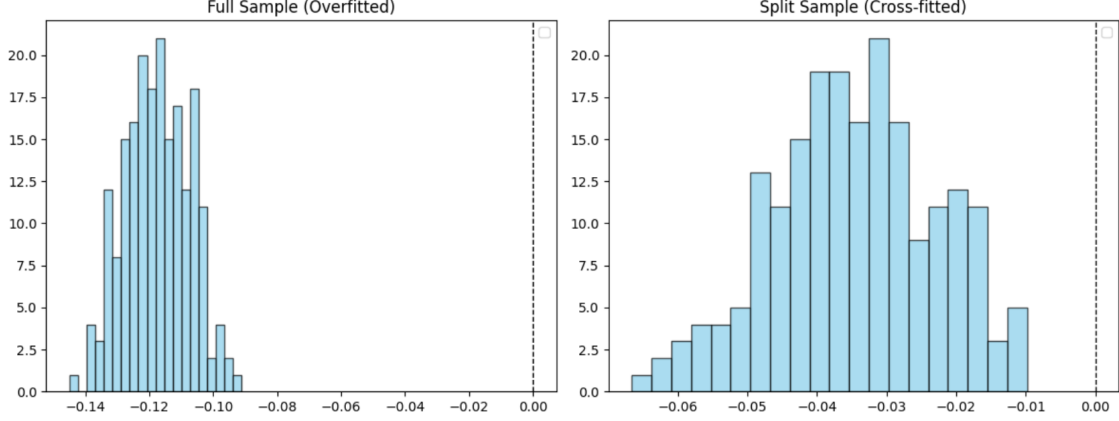


Figure 2: Effect of sample splitting on $\check{\theta}_0$, calculated on synthetic data; the line at 0 represents the perfect estimation.

DML Implementation Road Map

To practically implement the Debiased/Double Machine Learning methodology for the PLR model, we follow a structured four-step road map that integrates both cross-fitting and the Neyman-orthogonal score.

Step 1: Cross-fitting (Data Splitting)

First, it is crucial to divide the total sample of N observations into K distinct folds. For each fold k , the data is strictly separated into two parts: an "Auxiliary Sample" (denoted as I_k^c) which is used exclusively to train the machine learning algorithms, and a "Main Sample" (denoted as I_k) which is used exclusively to perform inference.

Step 2: Nuisance Parameter Estimation

Using the Auxiliary Sample (I_k^c), we estimate the complex relationships between the control variables X and the main variables Y and D . We can apply any suitable Machine Learning tool to obtain the conditional expectation functions:

$$\hat{l}_k^c(X) = E[Y|X] \quad (7)$$

$$\hat{m}_k^c(X) = E[D|X] \quad (8)$$

These estimates serve as our high-dimensional nuisance parameters.

In the formula, the $\hat{l}(X)$ estimate is present instead of $\hat{g}(X)$ because $\hat{g}(X)$ is not directly observable in the data. Thus, we use $\hat{l}(X)$, whose relationship with $g_0(X)$ is clear and explained by $g_0(X) = l_0(X) - \theta_0 m_0(X)$.

Step 3: Partialling Out (Residualization)

Once the ML models are trained on the auxiliary data, we apply them to the Main Sample (I_k). In

this step, we remove the "noise" or confounding variations explained by the controls X to obtain orthogonal residuals. The residualization is performed as follows:

$$\hat{U} = Y - \hat{l}_k(X) \quad (9)$$

$$\hat{V} = D - \hat{m}_k(X) \quad (10)$$

This process isolates the identifying variation needed to accurately estimate the causal effect.

Step 4: Solving the Neyman-Orthogonal Score

In the final step, we estimate the target parameter $\hat{\theta}_k$ using the orthogonalized residuals constructed in the Main Sample (I_k). To regain full efficiency and ensure that all available data is utilized, the final DML estimator is calculated as the average of the estimates across all K folds. The aggregated estimator is given by:

$$\tilde{\theta}_{DML} = \frac{1}{K} \sum_{k=1}^K \left(\frac{1}{n} \sum_{i \in I_k} \hat{V}_i D_i \right)^{-1} \frac{1}{n} \sum_{i \in I_k} \hat{V}_i \hat{U}_i \quad (11)$$

By rigorously following this road map, the resulting estimator successfully eliminates both regularization and overfitting biases, yielding consistent inference for the target parameter θ_0 .

Thus, the empirical moment formula representing the Neyman-Orthogonal score for the PLR model is defined as:

$$\psi(W; \theta, \eta) := \{Y - l(X) - \theta(D - m(X))\}(D - m(X)) \quad (12)$$

with the nuisance parameter defined as $\eta = (l, m)$, and $l_0(X) = E[Y|X] = m_0(X)\theta_0 + g_0(X)$ as seen before. Consequently, the orthogonal score is $\{Y - l(X) - \theta(D - m(X))\}$.

Machine Learning Models

Only two machine learning models have been chosen to calculate the nuisance parameters: Lasso and Random Forest. The main reason is to keep the computation from being too expensive, and the second reason is that the Random Forest model is already sufficient to reach a good estimate of θ_0 . Lasso has been chosen to show how much improvement we gain in the estimate when transitioning from a less flexible model (Lasso) to a much more flexible one (Random Forest).

For the Lasso model, the tuning parameters (alpha) have been decided using K -fold cross-validation for both the estimation of l_0 and m_0 . Instead, the tuning parameters for the Random Forest have been decided by an empirical rule based on observing the results, again for computational cost reasons. Regardless, the effect of the tuning parameters for this model was marginal on the final estimate.

Both the Lasso and Random Forest models have been used to estimate θ_0 using K -fold cross-fitting.

CATE Analysis

The PLR is the simplest way to apply the DML framework, and the central assumption of the model is that the ATE of interest is constant for the entire population. This is surely a strong assumption, but the DML framework can still be applied to cases where we relax this assumption. Therefore, to provide an idea of how this can be implemented, I divided the main sample based on one of the control variables and studied the behavior of the ATE conditional on this variable. In other words, I performed the so-called "subgroup analysis" or "Group Average Treatment Effect" (GATE), which is one of the most basic ways to study the Conditional Average Treatment Effects (CATE) based on this variable.

To choose the conditional variable for the CATE, the criterion used in this study was based on the most evenly distributed variable in the sample, the one with the smallest difference between the median and the mean, which in our case is age. The study divides the sample based on the median age of 40 years. The sample was split into two different subsamples for age ≤ 40 and age > 40 , and the subsequent estimations conditional on this variable have been categorized as "young" and "old", respectively.

Using the estimated GATE for the young and old subsamples, the precision-weighted ATE was calculated. This estimate was then compared with the ATE estimation performed initially on the whole sample.

To do this, the study first uses the sizes of the two subgroups and the number of treated subjects in each group to calculate the probability of treatment:

$$p_{\text{young}} = \frac{1}{N_{\text{young}}} \sum_{i \in \text{young}} D_i$$
$$p_{\text{old}} = \frac{1}{N_{\text{old}}} \sum_{i \in \text{old}} D_i$$

Then, the variance of the treatment in the two subgroups was calculated using this formula:

$$\text{Var}_{\text{young}} = p_{\text{young}}(1 - p_{\text{young}})$$
$$\text{Var}_{\text{old}} = p_{\text{old}}(1 - p_{\text{old}})$$

This formula is used because it represents the variance of a binary variable. Using the variance, we calculated the weights:

$$w_{\text{young}} = \frac{N_{\text{young}} \text{Var}_{\text{young}}}{N_{\text{young}} \text{Var}_{\text{young}} + N_{\text{old}} \text{Var}_{\text{old}}}$$

$$w_{\text{old}} = \frac{N_{\text{old}} \text{Var}_{\text{old}}}{N_{\text{young}} \text{Var}_{\text{young}} + N_{\text{old}} \text{Var}_{\text{old}}}$$

The last step involves multiplying the GATE estimates for the two groups by their respective weights:

$$\hat{\theta}_{\text{final}} = w_{\text{young}} \hat{\theta}_{\text{young}} + w_{\text{old}} \hat{\theta}_{\text{old}}$$

3 Data Description

Sample Overview and Selection Criteria

The empirical analysis utilizes cross-sectional data drawn from the 1991 Survey of Income and Program Participation (SIPP). The dataset comprises $N = 9,915$ observational units. The target population consists of household reference persons aged 25 to 64, including spouses if present. To ensure sample consistency and focus on active wage earners, households are included only if at least one member is employed and no individual is self-employed. This dataset is a standard benchmark in the literature, previously analyzed by Chernozhukov and Hansen (2004) and re-examined in Chernozhukov et al. (2018) and many other papers.

Variable Definitions and Causal Structure

The dataset is structured to evaluate the causal impact of 401(k) eligibility on accumulated household wealth within a treatment framework:

- **Outcome Variable (Y):** The outcome variable is net financial assets (`net_tfa`), defined as the total sum of IRA balances, 401(k) balances, checking accounts, U.S. savings bonds, interest-earning bank accounts, personal bonds, stocks, and mutual funds, minus non-mortgage debt.
- **Treatment Variable (D):** The policy treatment of interest is an indicator variable for 401(k) plan eligibility (`e401`), taking a value of 1 if an individual works for a firm that offers access to a 401(k) plan, and 0 otherwise.
- **Control Variables (X):** The vector of raw covariates includes key economic and demographic confounders: household income (`inc`), age (`age`), years of education (`educ`), family

size (fsize), marital status (marr), two-earner household status (twoearn), home ownership (hown), IRA participation (pira), and defined benefit pension status (db).

Identification and Confounding Considerations

Determining the causal effect of 401(k) eligibility is complicated because working for an employer offering a 401(k) plan is not randomly assigned. Following the identification strategy of Poterba et al. (1994a,b), 401(k) eligibility can be treated as exogenous only after conditioning on observables, most notably income and job choice characteristics. The core argument posits that around the time 401(k) plans were introduced, workers rarely selected jobs based on 401(k) availability. Furthermore, the dataset also includes a variable called 401(k) participation (p401), which could have been a candidate for the treatment variable, but e401 was preferred. Since p401 is a self-selected decision driven by unobservable individual traits such as financial discipline and risk tolerance, eligibility (e401) serves as a better treatment variable.

Summary Statistics

1991 SIPP Sample

	Y	D	X (Controls)									Others		
Stat	net_tfa	e401	age	inc	educ	fsize	marr	twoearn	pira	hown	db	nifa	tw	p401
Mean	18,051.5	0.37	41.1	37,200.6	13.2	2.9	0.60	0.38	0.24	0.64	0.27	13,928.6	63,816.8	0.26
Std	63,522.5	0.48	10.3	24,774.3	2.8	1.5	0.49	0.49	0.43	0.48	0.44	54,904.9	111,529.7	0.44
Min	-502,302.0	0.00	25.0	-2,652.0	1.0	1.0	0.00	0.00	0.00	0.00	0.00	0.0	-502,302.0	0.00
25%	-500.0	0.00	32.0	19,413.0	12.0	2.0	0.00	0.00	0.00	0.00	0.00	200.0	3,291.5	0.00
50%	1,499.0	0.00	40.0	31,476.0	12.0	3.0	1.00	0.00	0.00	1.00	0.00	1,635.0	25,100.0	0.00
75%	16,524.5	1.00	48.0	48,583.5	16.0	4.0	1.00	1.00	0.00	1.00	1.00	8,765.5	81,487.5	1.00
Max	1,536,798.0	1.00	64.0	242,124.0	18.0	13.0	1.00	1.00	1.00	1.00	1.00	1,430,298.0	2,029,910.0	1.00

4 Results

Estimated Average Treatment Effect (ATE)

The application of the Double/Debiased Machine Learning (DML) methodology to the 1991 SIPP dataset yields insightful estimates regarding the causal impact of 401(k) eligibility on net financial assets. As stated in the methodology section, the nuisance parameters were estimated using Lasso (a regularized linear model) and Random Forest (a highly flexible, non-linear model). To analyze the impact of the data-splitting process, the cross-fitting procedure was executed using 2, 5, and 10 folds. The aggregated point estimates for the Average Treatment Effect (ATE) are summarized in the table below:

Table 1: Estimated Average Treatment Effect (ATE) of 401(k) Eligibility

Cross-Fitting	Lasso Estimate (\$)	Random Forest Estimate (\$)
2-Fold	5,868.7	9,511.9
5-Fold	5,871.9	9,362.9
10-Fold	5,874.4	9,262.5

Analysis of the Results

The results obtained in this paper are consistent with those of Chernozhukov (2018), confirming that the procedure has been applied correctly and allowing us to focus on the findings.

The literature about 401(k)s in the past has estimated the same ATE, and the results lie between \$8,900 and \$9,300. Consequently, it is safe to say that the values from the Random Forest are much more precise than the Lasso ones, which is due to the fact that the Lasso model imposes a linear structure on the effect of the predictors on the target variable. Because the discrepancy between the treatment effect estimated via Lasso and the one obtained through Random Forest is so high, it is highly probable that the relationships among the confounding variables are complex and non-linear. Thus, the Random Forest is better equipped to absorb the structural noise and isolate the true causal parameter, and it appears that the estimation with 10-fold cross-fitting leads to the best result. The average effect of eligibility for the 401(k) program on net financial assets is \$9,262.

The Impact of Cross-Fitting Variations (K -Folds)

The choice of the data-splitting parameter (K -folds) impacts the final DML estimates differently depending on the machine learning model used.

For the Lasso estimator, the ATE remains remarkably stable across 2, 5, and 10 folds, showing only marginal fluctuations. This stability occurs because lower-complexity models require smaller sample sizes to converge, making them largely invariant to the auxiliary training sample size. Conversely, the Random Forest estimates exhibit a distinct downward trend, with the ATE dropping from \$9,511.9 (2-fold) to \$9,262.5 (10-fold). This reflects the data-hungry nature of highly flexible, non-parametric models. Under a 2-fold split, using only 50% of the sample for training increases out-of-sample prediction errors and bias. Increasing to 10 folds allocates 90% of the data to the auxiliary training sample, granting the Random Forest substantially more information to accurately map intricate confounding structures.

CATE and Subsequent Analysis

As demonstrated in the methodology section, the study has shown a basic version of CATE analysis with the DML sample. The framework was applied to both subsamples, considering only the

Random Forest with 10-fold estimation, as it is the better model of the two used in this analysis. The results for $\hat{\theta}$ for the old and young groups are shown below:

$$\hat{\theta}_{\text{young}} = \$6572 \quad \hat{\theta}_{\text{old}} = \$13029$$

Below, we can see the results for the sample mean of the treatment, the variance of the treatment, and the resulting weights for the two subgroups based on age.

Metric	Young (≤ 40)	Old (> 40)
GATE ($\hat{\theta}$)	6572.47	13028.80
Treatment Mean (p)	0.349	0.396
Treatment Variance	0.227	0.239
Precision Weight (w)	0.510	0.490
Final ATE Estimate	9735.66	

The estimated ATE obtained from the estimation on the two subgroups is different from the estimation of the ATE on the whole sample. The difference is equal to $9,735.66\$ - 9,262.5\$ = 473.16\$$

5 Conclusion

This paper demonstrated the practical implementation of the Double/Debiased Machine Learning (DML) approach to overcome regularization and overfitting biases when estimating causal effects with high-dimensional confounders. Although the treatment effect analysis of 401(k) eligibility on net financial assets in this study is surely not exhaustive, the primary intent was to provide an accessible, informative entry point into this methodology and spark the reader’s interest in modern causal inference. There are several more specifications that could complicate the analysis of the treatment effect model, but the intent of this study is simply to show one of the possible ways to create a more robust and complex empirical analysis. Importantly, this framework extends far beyond the Partially Linear Regression (PLR) model used here, accommodating fully non-parametric setups, instrumental variables, and dynamic panel models. Furthermore, by leveraging the flexibility of advanced machine learning algorithms, researchers are not limited to traditional tabular covariates; modern applications can seamlessly integrate highly complex, unstructured data, such as text, audio, or images, as confounding nuisance parameters. Ultimately, DML provides empirical researchers with a rigorous and highly adaptable toolbox to draw valid causal conclusions in the era of big data.

References

Academic Papers

- Chernozhukov et al. (2018). *Double/Debiased Machine Learning for Treatment and Structural Parameters*.
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Online Resources

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